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Summar Imaan Arshad

Abstract

My report goes over the creation of a medium-sized computer network that is designed to support 50 hosts across multiple locations. The network includes both wired and wireless connections, which was built using a software called Cisco Packet Tracer. CIDR (Classless Inter-Domain Routing) was used to manage IP addressing more efficiently on the internet and reduce any wasted address space. The dynamic routing between subnets was set up using RIP version 2 (which supports CIDR and makes routing between subnets easier). The network design is focused on reliability, keeping low costs, future growth, and making sure the network stays available if a link happens to fail.

Introduction

NorthernTech Systems Ltd is a small organisation with multiple sites that requires a reliable and scalable network to support its daily operations. The company operates from four office locations and one workshop, all of which require both wired and wireless network access for staff, servers, and shared resources. As NothernTech Systems Ltd continues to expand, the network must be able to support up to 50 hosts whilst allowing smooth communication between the sites, providing basic fault tolerance, and remaining easy to manage.

My project will focus on designing and building a medium-sized enterprise network to meet the organisation's requirements. The network design will use CIDR to efficiently look up IP addresses, while dynamic routing using the RIP version 2 will be implemented to allow any communication between the different subnets. The hosts will be configured to receive IP addresses automatically through the DHCP. The network devices will be managed securely using SSH for remote access. The software that I will be using, Cisco Packet Tracer, will be used to design, implement, and test the network, as it allows realistic simulation of routers, switches, servers, and end devices.

The main take away for my project is that all sites have reliable infrastructure and that the network design prioritises low cost while maintaining performance and reliability. My report will cover the network design, configuration process, verification through the testing process, and an evaluation of my created network's scalability and reliability.

Aims and Objectives

Aim:

The aim will be to design and build a secure, scalable medium-sized network. It will have to support up to 50 hosts across multiple locations using both wired and wireless connectivity.

Objectives:

1. Develop an IP address plan using CIDR. It should efficiently support 50 hosts.
2. Design a multi-site network layout with router-to-router connection.
3. Implement RIP version 2 to enable dynamic routing between subnets.
4. Configure DHCP to automate IP address location for the hosts.
5. Enable SSH to provide secure remote management of the network devices.
6. To test and verify end-to-end connection between hosts at different locations.

Network Design

The suggested network design connected four different locations using routers that were linked together with serial connections. Every location represented an office or workshop and included its own router, a Layer 2 switch, user PCs, and servers. Wireless access was provided at selected sites using wireless routers or access points that were connected to the local switch. This allowed laptops and other devices to connect to the network.

Cisco 2811 routers were chosen as they are suitable for the organisation I designed the network for, which is a medium-sized network. They supported serial connections and worked very well with RIP version 2. Cisco 2960 switches were used at each site to provide affordable Layer 2 switching, with enough ports to support multiple devices. Servers were included to run network services such as DHCP, while access points were used to add wireless access without making the cabling complicated. I believe that this design supports future growth by allowing additional subnets or locations to be added, whilst also maintaining a good balance between cost and performance.

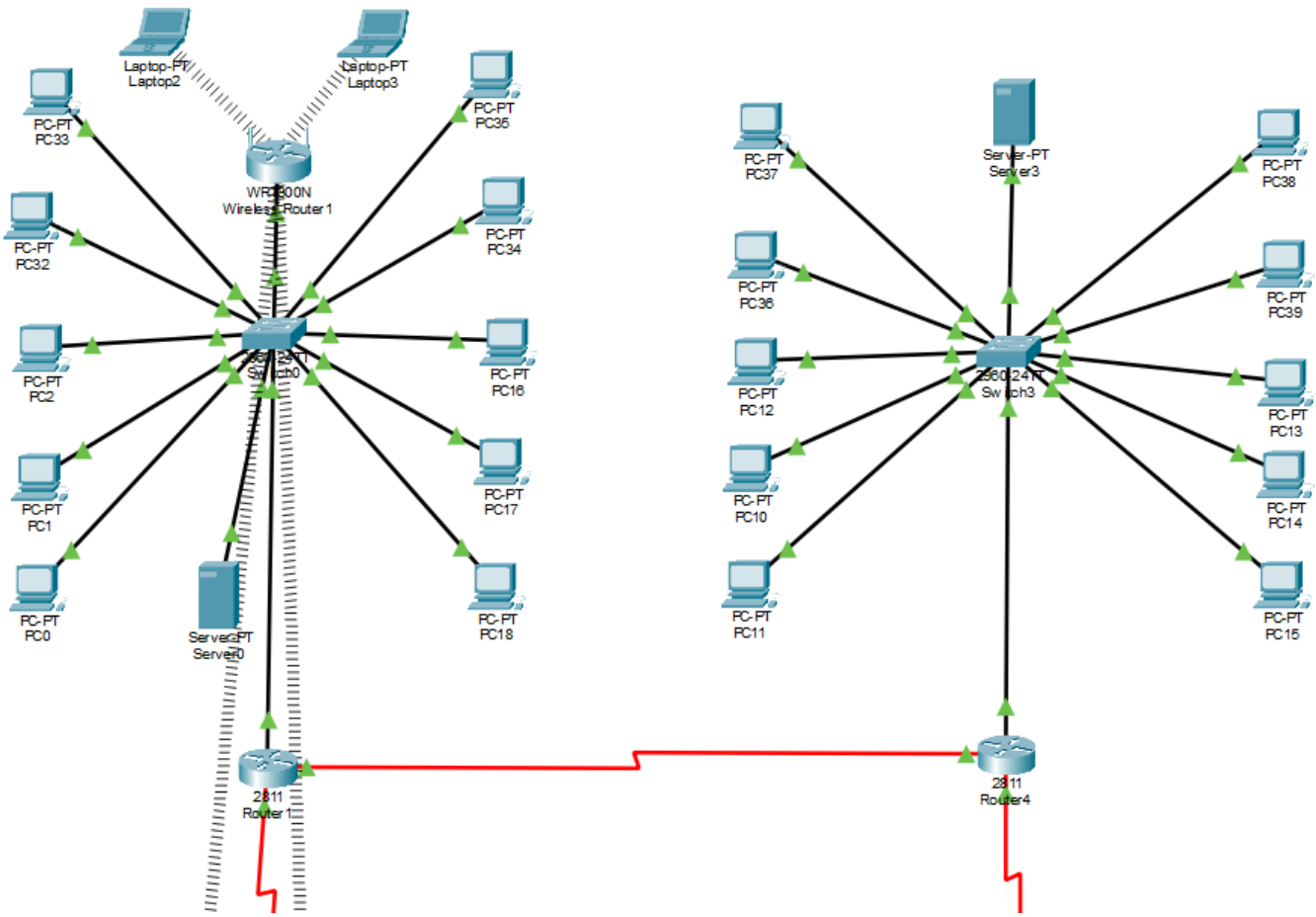
IP Addressing and Subnetting

A CIDR-based IP addressing scheme was developed to share IP addresses efficiently, while also allowing the network to grow for the future. IP addresses are automatically assigned to devices using DHCP services that run on the servers. Each site was allocated a '/27' subnet (255.255.255.224). This provides up to 30 usable host addresses. It also easily meets the requirement of around 10 hosts per location, whilst leaving room for any growth. Point-to-point serial connections between the routers use '/30' subnets to reduce wasted IP addresses, whilst still supporting reliable communication between the routers. I believe that I have met the first and second objectives, as the addressing approach meets what they need.

The subnet allocations for each site:

- **Site 1:** 192.168.10.0/27
- **Site 2:** 192.168.20.0/27
- **Site 3:** 192.168.30.0/27
- **Site 4:** 192.168.40.0/27

Figure 1



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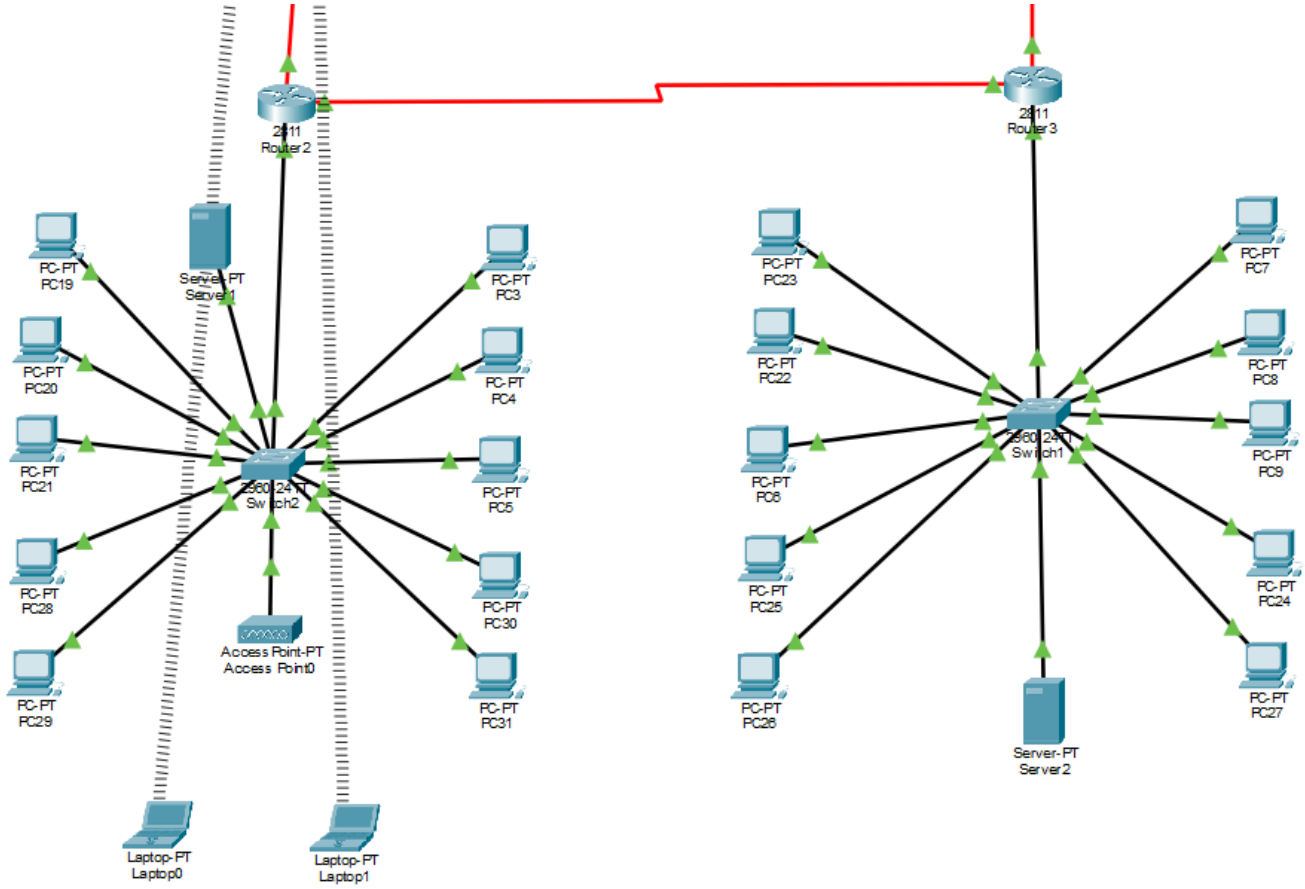

Figure 1 shows an image of the topology

Routing Protocol Selection

RIP version 2 was configured on all the routers to enable dynamic routing between the different subnets. It has been chosen because it is extremely easy to set up, and it also works well for small/medium networks, and also supports CIDR. The auto-summary was disabled to avoid routing problems and make sure that routers advertise the correct information of the network. This helps traffic reach the correct destination across all the network locations.

Figure 2

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:01, Serial0/3/0
R       10.0.0.8/30 [120/1] via 10.0.0.13, 00:00:20, Serial0/3/1
C       10.0.0.12/30 is directly connected, Serial0/3/1
L       10.0.0.14/32 is directly connected, Serial0/3/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/27 is directly connected, FastEthernet0/0
L       192.168.10.1/32 is directly connected, FastEthernet0/0
     192.168.20.0/27 is subnetted, 1 subnets
R       192.168.20.0/27 [120/1] via 10.0.0.2, 00:00:01, Serial0/3/0
     192.168.30.0/27 is subnetted, 1 subnets
```

Figure 2 shows an image of the learnt routes that are active on router one.

DHCP Configuration

For the DHCP configuration, it was configured on the servers within each network and connected to the routers so that all the connected PCs can automatically receive IP addresses. Using DHCP has reduced the need for manual configuration, helps prevent any IP address conflicts, and makes the network a lot easier to manage. Each DHCP pool matches its assigned subnet and provides hosts with the correct default gateway and the DNS information. This allows new devices to join the network immediately without requiring any manual setup.

The DHCP configuration follows the guidelines described in the Cisco IOS DHCP Configuration Guide, which made it easy to set up and no problems as I did it the recommended way. Using DHCP also makes the network easier to manage. This is because IP addresses and network settings can be controlled rather than being configured on each device.

Figure 3

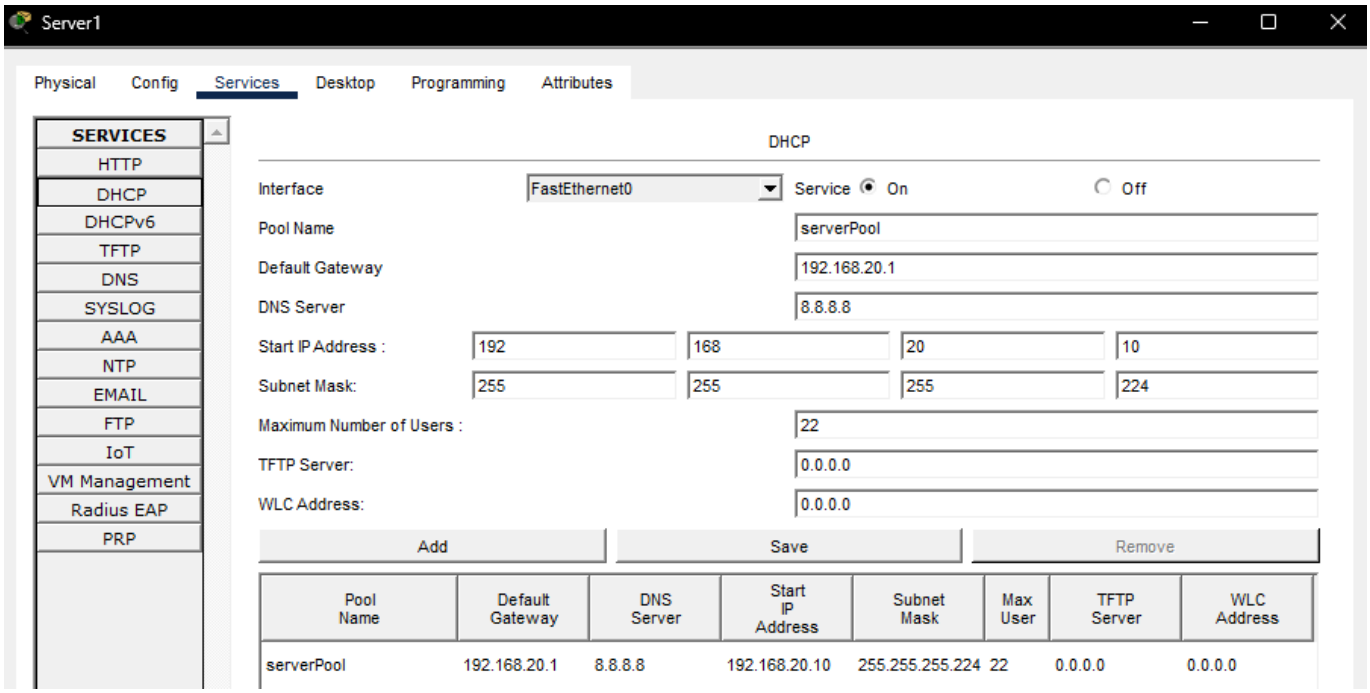

Figure 3 shows an image of the DCHP server configuration on site one.

Secure Shell (SSH)

SSH was configured on one router to allow secure remote management. It was chosen instead of the older protocols, such as Telnet, because it encrypts data, which helps prevent login credentials and configuration information being intercepted. Local user accounts and RSA keys were set up to make sure that only the authorised users would be able to access the device remotely.

Figure 4

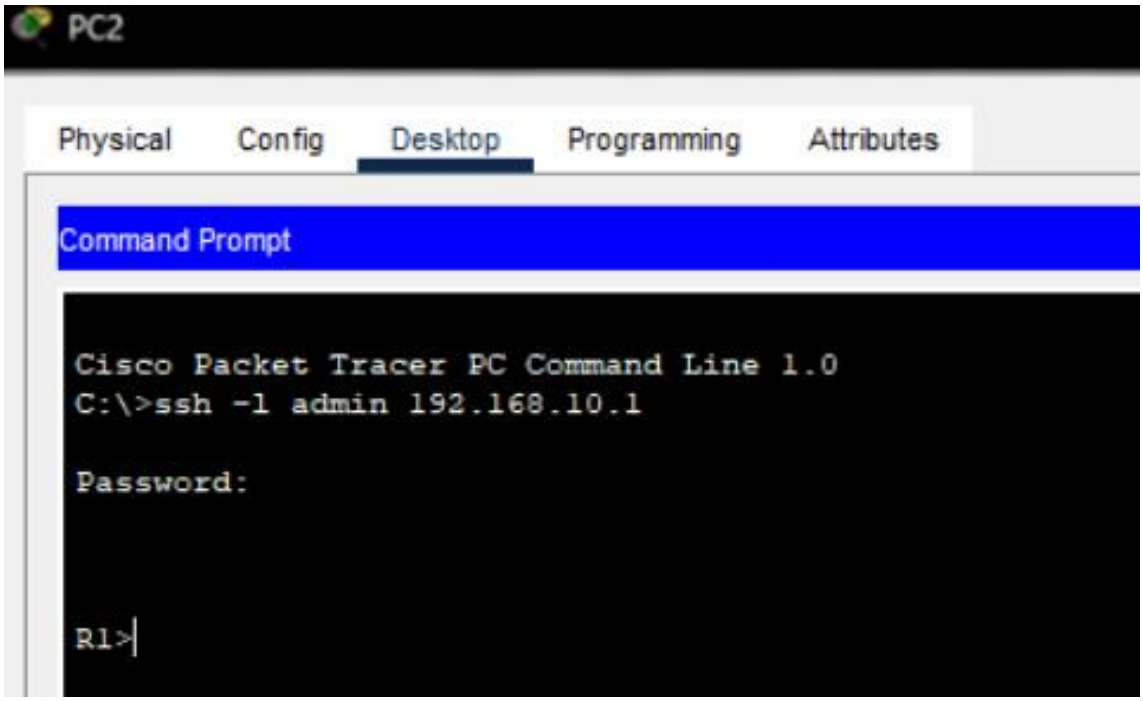

Figure 4 is an image showing a screenshot taken on a PC connected to the router on site one.

Network Configuration Tests

Figure 6

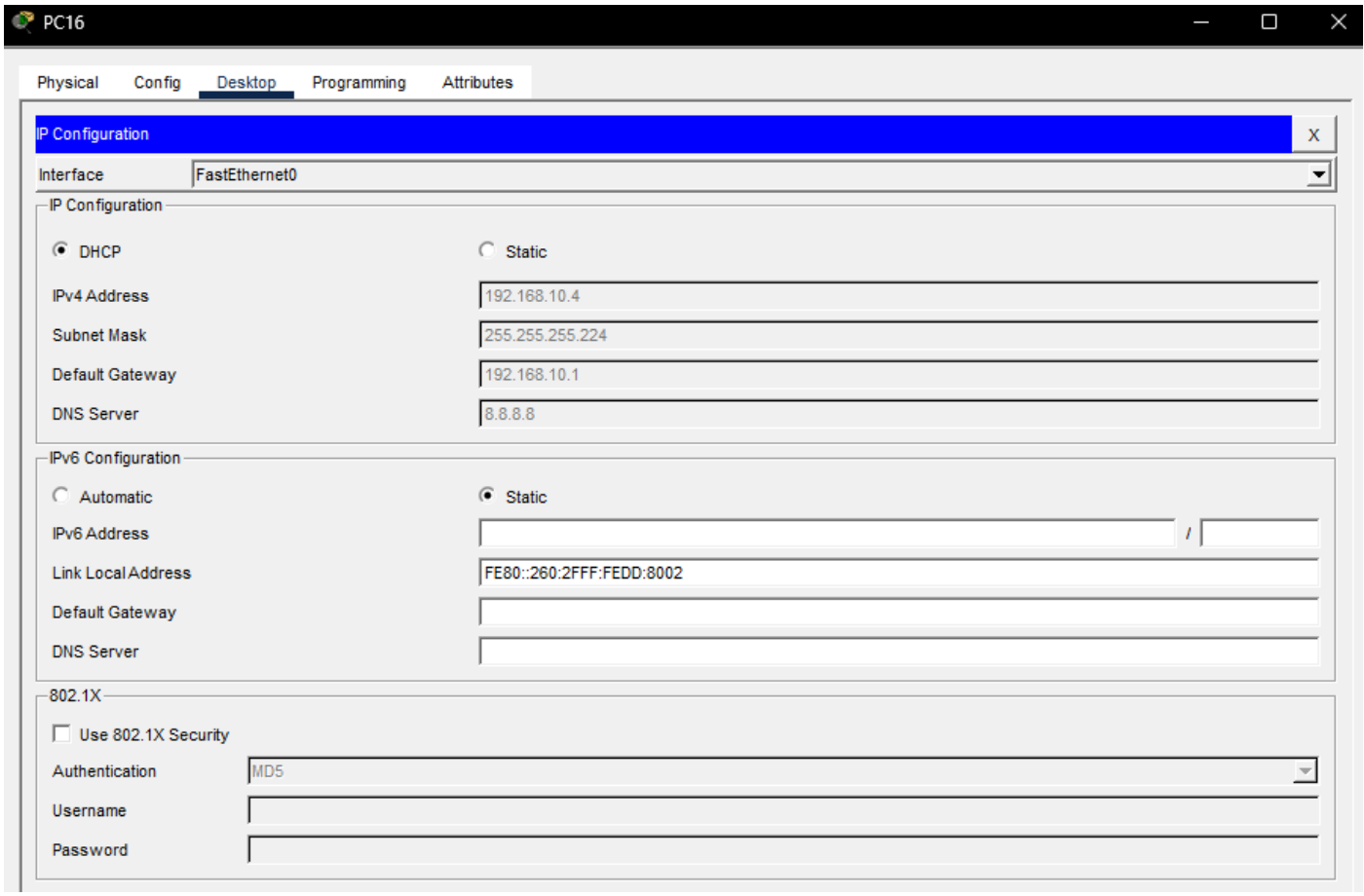

Figure 3 and 6 shows images of a PC in site one.

DHCP Verification & Address Allocation Test

In the ‘Network Configuration Tests’ section, it shows that the DHCP service is enabled on the server at site 1. This is configured with a pool of IP addresses that match the ‘192.168.10.0/27’ subnet. Figure 6 also show a client device successfully receiving an IPv4 address along with key network details, such as the subnet mask, default gateway, and DNS server. All of this confirms that I have set up the DHCP service correctly.

Using DHCP makes it simple for the network administration, and prevents IP addresses from being tampered with, by making sure each device is assigned at a unique address. I believe that this section supports the second objective as it demonstrates automated and scalable IP address allocation.

Figure 7

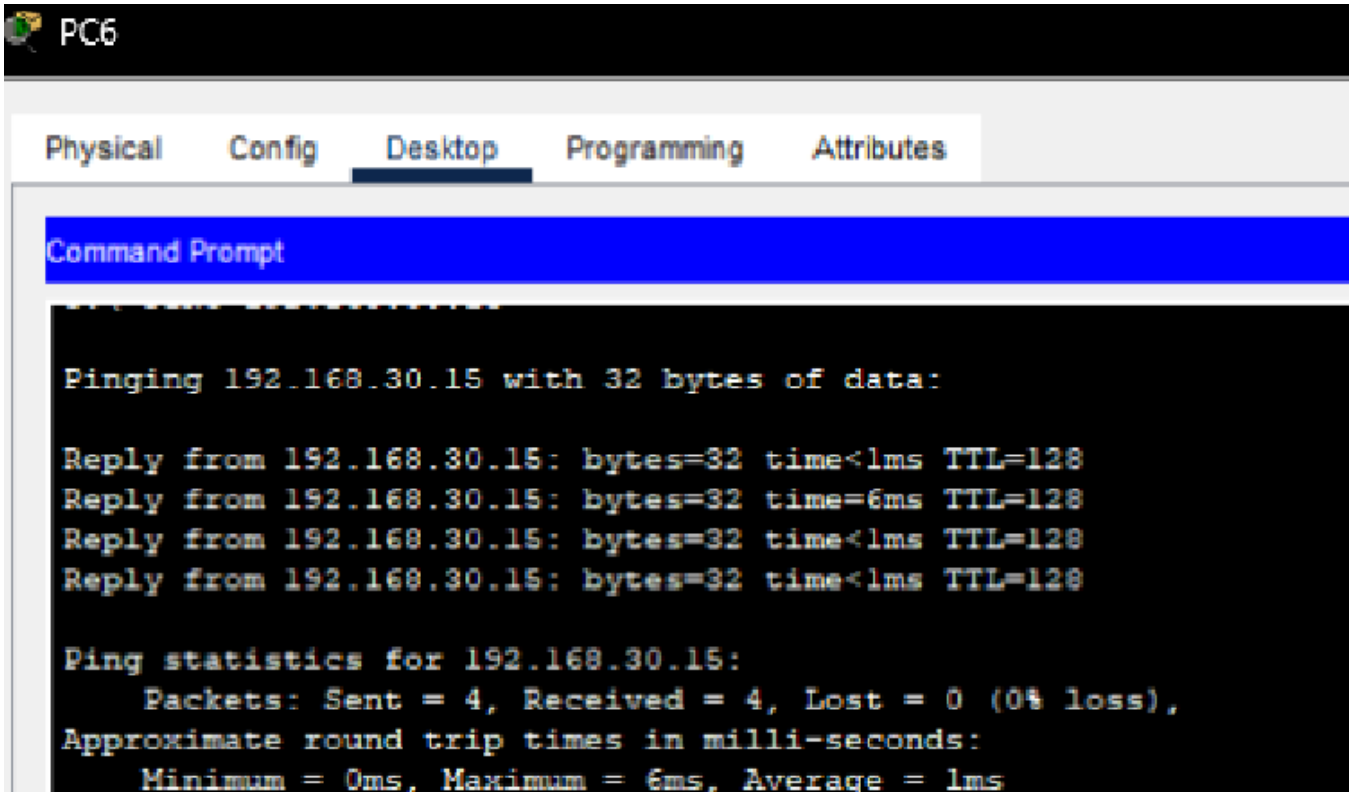

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Figure 8

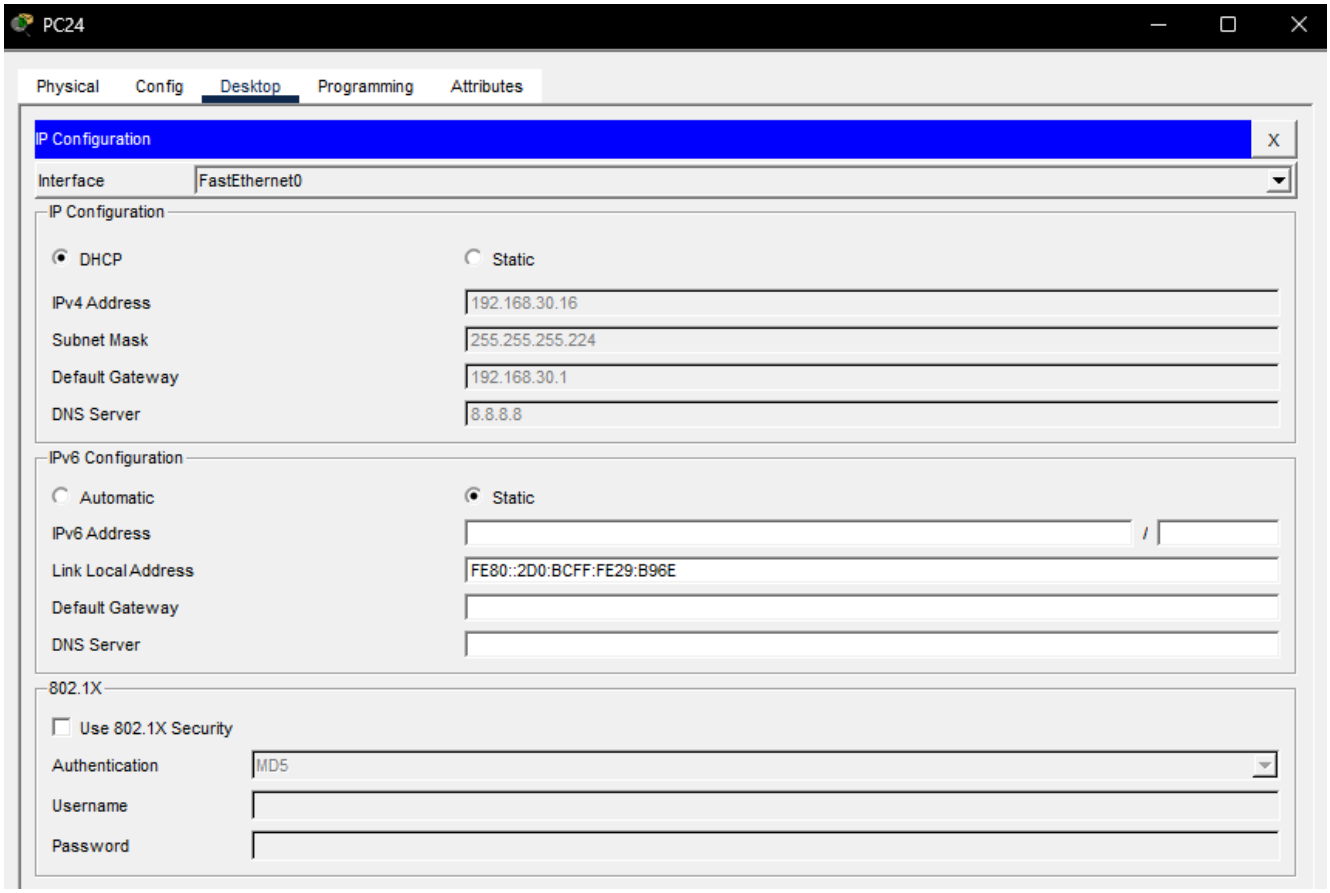

Figure 7 and 8 shows two screenshots of two different PCs in site three.

Local Subnet Connectivity Test

The connectivity within a single site was tested to check that the computers would be able to communicate with each other. This was done by sending ping requests from one computer to another computer in the same location. The successful replies in figure 9 show that the network connection and IP address configuration are working correctly at that site. I believe that this section supports the fourth objective, as it confirms end-to-end connectivity.

Figure 9

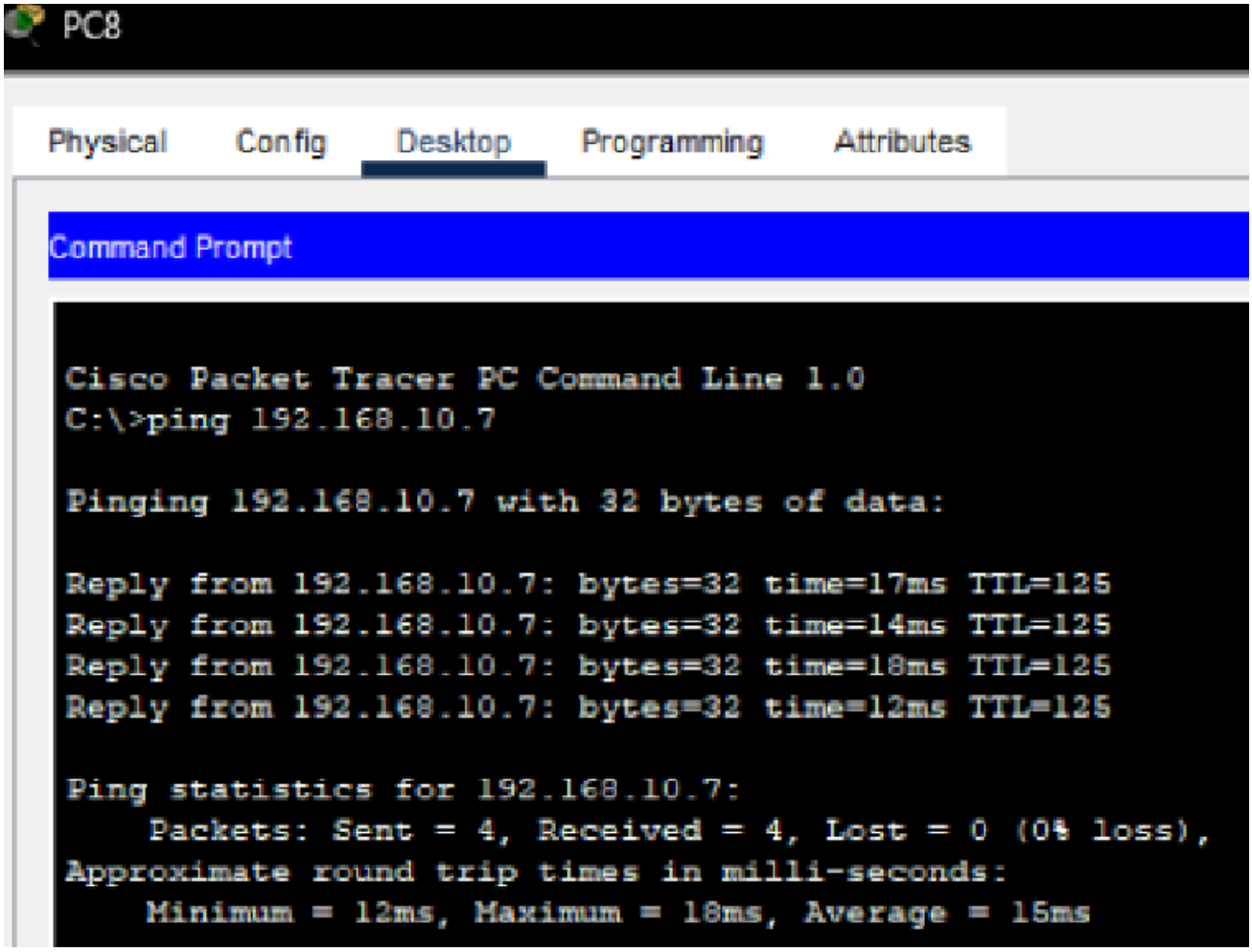

Figure 9 shows a screenshot taken on a PC in site three.

Wide Network Subnet Connectivity Test

The ‘ipconfig’ output shows that the source PC is on the ‘192.168.10.0/27’ subnet, whilst the destination PC is on a different subnet. The successful ICMP replies confirm that the traffic is being routed correctly between the subnets. This test shows that RIP version 2 is working as expected and that all the sites will be able to communicate with each other across the WAN. I believe that the sixth objective has been met, as it confirms the connection between hosts at different locations.

Figure 10

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up


Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:28, Serial0/3/0
R       10.0.0.8/30 [120/1] via 10.0.0.13, 00:00:08, Serial0/3/1
C       10.0.0.12/30 is directly connected, Serial0/3/1
L       10.0.0.14/32 is directly connected, Serial0/3/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/27 is directly connected, FastEthernet0/0
L       192.168.10.1/32 is directly connected, FastEthernet0/0
     192.168.20.0/27 is subnetted, 1 subnets
R       192.168.20.0/27 [120/1] via 10.0.0.2, 00:00:28, Serial0/3/0
 --More--
```

Figure 11

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
     192.168.40.0/27 is subnetted, 1 subnets

Router>nd
Translating "nd"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 14 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0       22
  Serial0/3/0           22
  Serial0/3/1           22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
            10.0.0.0
            192.168.0.0
            192.168.10.0
Passive Interface(s):
Routing Information Sources:
            Gateway         Distance      Last Update
            10.0.0.2             120      00:00:25
            10.0.0.13            120      00:00:05
 --More--
```

Figure 10 and 11 shows images of a screenshot of CLI in router one.

Dynamic Routing Verification (RIP V2)

RIP version 2 was checked to make sure that it was working properly using the 'show IP route' and the 'show IP protocols' commands. In figure 10, it shows the 'show IP route' output, which routes are marked with the letter 'R'. This indicates that they were learned dynamically through RIP version 2, rather than being configured manually. The 'show IP protocols' output (figure 11), shows that RIP version 2 is running on the router and showing the correct networks.

RIP version 2 was configured with auto-summary disabled, as I have explained before, so that the routes accurately use CIDR prefixes. This helps prevent the routing problems, such as loops or unreachable networks, and confirms that dynamic routing is working across the WAN. It also confirms that the network can support future growth.

RIP version 2 supports classless routing, and it is suitable for small/medium networks. The use of WAN links reduces single points of failure and improves the network's availability. I believe that this section supports my third objective, as it shows that the dynamic routing between sites is functioning correctly.

Figure 12

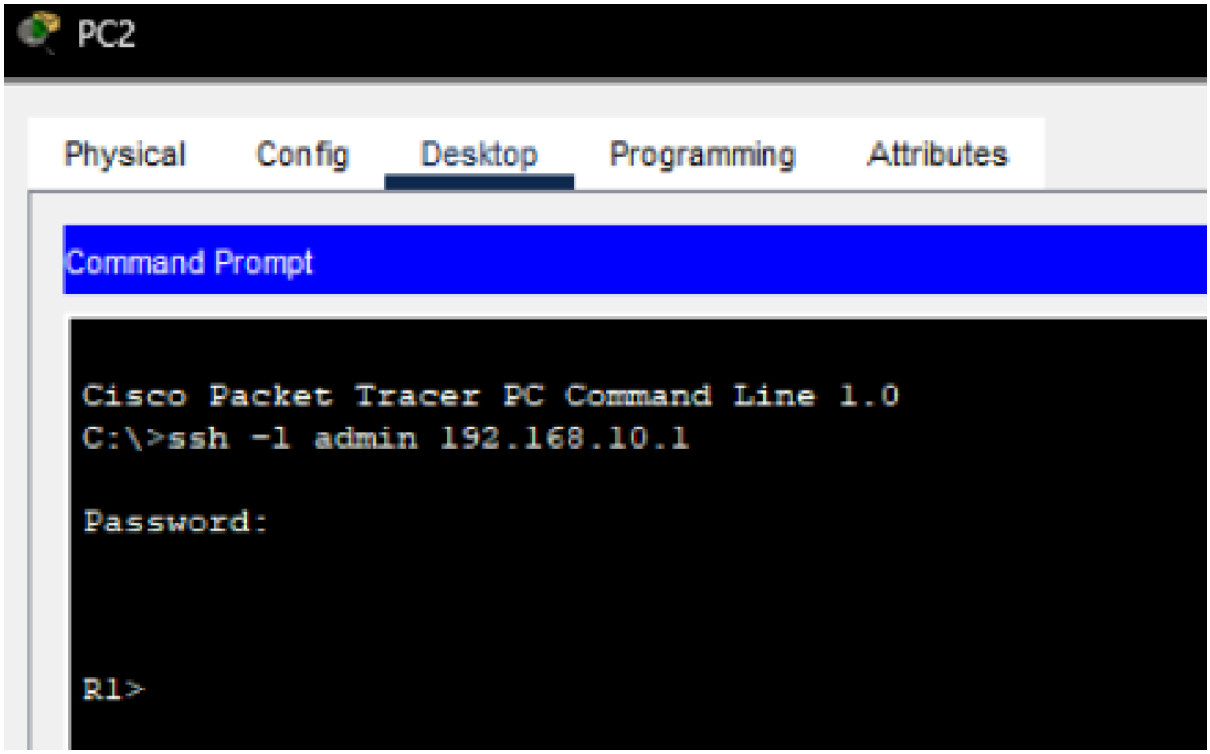

Figure 12 shows a screenshot of 'PC2' in site one.

SSH Remote Management Test

Remote management of the router was tested by configuring SSH on Router R1 and connecting it to a computer in a different location. As you can see in figure 12, the SSH login attempt was successful using the user authentication configured on Router R1. This shows that the remote access is working correctly.

SSH was used because it provides encrypted management access, which helps protect login credentials and configuration data from being intercepted. During testing, the first ICMP ping request had occasionally failed, but only due to the ARP resolution, since the device needed to learn the MAC address of the next hop. Once the ARP table was updated, the following ping requests were successful. This behaviour tends to be normal and expected in IP-based networks.

I believe that this section meets the fifth objective, as it allows a secure remote administration of network devices.

Network Evaluation and Discussion

The redundancy was addressed by using multiple serial links between the routers arranged in a square topology. This design removes a single point of failure within the WAN, as traffic can still be routed through alternative paths if one link fails. RIP version 2 supports this by automatically using another route when a connection happens to go down. Even though adding extra links and routers increases the overall size of the network, this trade-off is justified, since reliability is a requirement in organisations and helps minimise any disruption between the sites.

The scalability was also considered in the network design. Each site was assigned a '/27' subnet, which provides up to 30 usable host addresses. This definitely exceeds the current requirement of only 50 hosts; however, it allows future growth without needing to re-design the IP addressing scheme. Any additional wired or wireless devices can be easily added, and new sites can be included by assigning new subnets through RIP version 2. DHCP also further improves the scalability by automatically assigning IP addresses, and reducing the amount of manual configuration needed as the network continues to grow.

If you are looking at it from a cost perspective, I would say that it is quite cost-effective. RIP version 2 was chosen for this sole reason, because it is easy to configure and avoids any complexity associated with more advanced routing protocols. Whilst RIP version 2 has its' limitations, it is suitable for the size of the network I have created. If the organisation expands any further, upgrading to a protocol, such as 'OSPF' (Open Shortest Path First), should be considered for NothernTech Systems Ltd's future.

Looking into the performance, separating each site into its own subnet helps reduce unnecessary broadcast traffic. In the future, VLAN (Virtual Local Area Network) could be introduced to further improve the traffic management within the different sites. The security was also improved by using SSH for remote router management, which encrypts management traffic and protects the login credentials from being intercepted.

I believe that the network design I have created meets the client's requirements and provides a good balance between cost, redundancy, scalability, simplicity, and security. I think it is suitable for this organisation, as it is medium-sized, and can be used in the future when the organisation grows.

Conclusion

In conclusion, my report goes through the successful design and implementation of using wired and wireless connections to support 50 hosts across different locations. A CIDR-based IP addressing scheme was used to make efficient use of IP addresses, whilst the DHCP was configured to automatically assign network settings to hosts and reduce the administrative workload. The dynamic routing was implemented using RIP version 2 with the auto-summary disabled, which allows all the subnets to be used correctly and makes sure it has reliable communication across the network.

The network was tested through the DHCP verification, inter-subnet connectivity checks, and making secure remote management using SSH. These tests confirmed that the network meets the functional, scalability, and security requirements that are outlined in the report. If NorthernTech Systems Ltd would like to expand further, the network could be improved by introducing VLAN or more advanced routing protocols. However, the current design is more than suitable for meeting the client's needs.

Appendix

Figure 13

Figure 13 shows a screenshot of ‘Router1’ in CLI. This provides basic local access control.

References

Cisco Systems (no date) DHCP – Configuring the Cisco IOS DHCP server. Available at: https://www.cisco.com/en/US/docs/ios/12_4t/ip_addr/configuration/guide/htdhcpsv.html (Accessed: 2 January 2026).

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SPOTO (no date) Understanding RIPv2 and RIPng: a comprehensive guide. Available at: https://cciedump.spoto.net/newblog/understanding-ripv2-and-ripng:-a-comprehensive-guide.html (Accessed: 10 January 2026).